

MOURYA BIRRU

Portfolio Website - <https://mourya-birru.vercel.app>

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LeetCode | **Linkedin** | **GitHub** | **HackerRank**

Completed over 300 LeetCode problems and earned multiple HackerRank certifications in Python, Java, SQL, and problem-solving. MongoDB Certified Associate Developer with hands-on experience in developing full-stack projects using Java, Node.js, Firebase, and MongoDB. Currently interning in the fields of generative AI and cybersecurity. Strong interest in building scalable backend systems, AI/ML-driven applications, and intelligent automation solutions.

EDUCATION

Geethanjali College of Engineering and Technology

Computer Science and Engineering Bachelors

CGPA: 8.25

Hyderabad, India

September 2023 - Present

Resonance Junior College

MPC Intermediate

Percentage: 84.9%

Hyderabad, India

2022 - 2023

Rajiv Gandhi University of Knowledge Technologies

PUC - 1

Basar, India

2021 - 2022

Pallavi Model School

Degree in Schooling

Percentage: 87.8%

Hyderabad, India

2015 - 2021

EXPERIENCE

Avasan Chakra | Software Engineer Intern

Hybrid | April 2026 - Present

- Engineered remote software development projects, applying algorithmic problem-solving to build sustainable technology solutions.
- Contributed to the design and implementation of eco-friendly technology initiatives specifically targeted for high-rise buildings.
- Collaborated directly with the Chief Technical Officer to deliver innovative solutions, ensuring clarity and precision in code execution.

Google | Student Ambassador

July 2025 - December 2025

- Selected as a Student Ambassador for programs led by Google, representing and promoting tech initiatives within the student community.
- Organized and supported workshops, events, and awareness activities to encourage student participation in technology programs.
- Built strong peer networks and increased engagement through campus outreach and online promotion.

SKILLS

Programming Languages:	C, Java, Python, R, HTML, CSS, JavaScript, Node.js, TypeScript
Libraries/Frameworks:	Tensor Flow, PyTorch, Scikit-learn, Keras, OpenCV, NumPy, Pandas, Matplotlib, Seaborn, Statsmodels, React.js, Bootstrap, Tailwind CSS, Express.js
Tools / Platforms:	Git, VS Code, PowerBI, Microsoft Office 365, GitHub, Jupyter Notebook, Google Console, Tableau, Figma, Firebase
Databases:	SQL, MongoDB

PROJECTS / OPEN-SOURCE

Hobby OS (x86 Kernel) | [Link](#)

C, x86 Assembly, GNU Assembler, QEMU

A custom-built, low-level x86 operating system kernel developed from scratch.

- Implemented foundational kernel structures including the Global Descriptor Table (GDT), Interrupt Descriptor Table (IDT), and memory paging.
- Architected keyboard driver and hardware-level interface protocols to enable functional user-mode input processing.

- Leveraged Assembly for bootloader bootstrapping and low-level CPU state management, ensuring stable transition to protected mode.
Skills: Systems Programming, Kernel Development, Low-level Optimization, Hardware Architecture

Air Canvas | [Link](#)

Python, OpenCV, NumPy, Computer Vision

An interactive gesture-based painting application using real-time hand tracking.

- Implemented color detection and masking algorithms in OpenCV to track specific colored objects for dynamic drawing inputs.
- Developed frame-by-frame spatial coordinate mapping to translate real-world hand gestures into digital strokes on a virtual canvas.
- Optimized image processing pipelines to minimize latency, ensuring smooth and responsive interaction for a real-time user experience.

Skills: Computer Vision, Real-time Image Processing, Human-Computer Interaction

Smart Attendance System | [Link](#)

Python, OpenCV, Computer Vision, Deep Learning, SQL

A real-time biometric attendance solution designed to automate classroom tracking.

- Implemented facial detection and recognition pipelines using OpenCV and deep learning models to identify students with high accuracy.
- Built a robust registration and training workflow to map unique student identifiers to facial features stored in a relational database.
- Developed an automated logging system that captures timestamps and attendance records, eliminating manual intervention and proxy issues.

Skills: Computer Vision, Real-time Processing, Biometric Authentication, Data Modeling

Career Compass | [Link](#)

TypeScript, Firebase, React, Tailwind CSS, Next.js, Genkit

A full-stack platform for personalized career guidance.

- Implemented Genkit-powered AI for resume analysis, skill mapping, and certification suggestions.
- Built dynamic UI with Tailwind, shadcn/ui, and Radix UI.
- Used Firebase for real-time storage and user authentication.

Skills: Full-stack development, Firebase Auth, AI-integration

EatIQ | [Link](#)

TypeScript, N8N, React, Tailwind CSS, Next.js

Developed EatIQ, a web app that analyzes food images and shows nutrition details using AI.

- Built the frontend using React, TypeScript, and Tailwind CSS with a clean and responsive design.
- Integrated AI workflows to process images and generate calorie and nutrient information automatically.

Skills: React, Node.js, AI integration, n8n, Web Development

MEMBERSHIPS

IEEE Computer Society | Student Member

September 2024 - January 2026

Indian Society for Technical Education | Student Member

February 2024 - Present

CERTIFICATIONS

- Networking Technician - **Cisco**
- MongoDB Associate Developer (NodeJS) - **MongoDB**
- Data Science and Machine Learning BootCamp with R - **Udemy**
- Machine Learning in Python - **365 Data Science**
- The Complete Full-Stack Web Development Bootcamp - **Udemy**
- Java - **HackerRank**

ACHIEVEMENTS

- Secured an AIR 9602 in GATE 2026 CSE
- Ranked 542 Globally and 149 Nationally in IEEE Extreme 18.0
- Secured Rank 1 in the Smart Interviews GCET-2027-R Coding Challenge among peers across the college